1. Idea Presentation

Due: 24 October 2026 | Weightage: 7% | Group | Submission Only | 27 Points |

Each group is required to present its proposed project idea, clearly explaining the motivation, identified research gap, proposed solution, methodology, and expected impact. The presentation must contain **exactly 5 slides**, organized as follows:

1. **Slide 1 – Motivation and Literature Gap:** Briefly discuss existing research, identify high-level limitations or gaps, and explain why addressing these gaps is important. (Note: A detailed literature comparison is not required at this stage.)
2. **Slide 2 – Proposed Solution and Novelty:** Explain your proposed solution, the problem it addresses, and how it differs from or improves upon existing approaches.
3. **Slide 3 – Dataset and Methodology:** Identify the dataset(s) you plan to use and explain your proposed methodology, including the main technical approach.
4. **Slide 4 – Probable Results:** Describe the expected outcomes, relevant evaluation metrics, and how you anticipate your approach will perform against existing baselines.
5. **Slide 5 – Impact:** Explain the potential significance of your work, who will benefit from it, and why the proposed solution matters.

Presentation Guidelines

- Each group will have **12 minutes for the presentation**, including **2 minutes of Q&A** with the professor.
- **Only one group member may present.** Other members must be present and answer Qs.
- Presentations exceeding the **time limit** will receive a grading penalty.
- Slides should be concise, visually clear, and avoid excessive text.
- Each group must submit its presentation in advance in **PDF or PPTX format**.
- Presentations will be delivered exclusively using the slides submitted on LMS, displayed from the instructor's computer. Personal laptops or updated versions of slides will not be used.

Rubric:

Criteria	Not Yet (0)	Fair (1)	Good (2)	Great (3)
Motivation & Gap	Missing, unclear, or fails to address existing work.	Vague motivation or superficial/weak gap identification.	Clear, relevant motivation and well-stated literature gap.	Convincing motivation with well-identified gaps and significance.
Solution & Novelty	Missing, unrelated, or lacks any novelty.	Vague solution with incremental or unclear novelty.	Clearly addresses the gap with discernible improvements.	Well-justified, highly novel, and clearly differentiated approach.
Dataset & Methodology	Missing dataset or fundamentally flawed/inappropriate methodology.	Incomplete design or loosely defined technical approach.	Appropriate dataset selected with a clear technical pipeline.	Technically sound, detailed, and well-justified methodology and data.
Probable Results	Missing or completely unrealistic outcome expectations.	Vague outcomes with missing or ill-suited evaluation metrics.	Reasonable outcomes grounded in relevant metrics.	Well-defined expected outcomes, clear metrics, and strong baselines.
Impact	Not discussed or fails to show practical/theoretical value.	Generic impact claim with no clear target beneficiaries.	Clearly identifies beneficiaries and practical significance.	Highly convincing, specific, and impactful contribution articulated.
Presentation	Unprepared, disorganized, or extremely difficult to follow.	Lacks clarity, focus, or engaging oral delivery.	Clear, well-structured, and easy-to-follow delivery.	Confident, articulate, professional, and highly effective delivery.
Slides	Incomplete, unreadable, or wrong format/slide count.	Cluttered, text-heavy, or poorly structured layout.	Exactly 5 slides, readable layout, concise content.	Exactly 5 visually effective, concise, and professional slides.
Q&A	Unable to answer or group members absent/unresponsive.	Partial, hesitant, or unclear responses to questions.	Mostly accurate, clear responses from the team.	Accurate, confident, and complete responses by team members.
Pacing / Length	>12 minutes (severely mismanaged timing).	>10 minutes for presentation (exceeds talk limit).	Within 10 minutes with minor pacing issues.	Completed strictly within 10 mins (+ 2 mins Q&A).

Scoring: *Each of the 9 criteria is worth 3 points, giving a maximum of 27 points.*

Weighted Score: *(Points Earned / 27) × 7*

2. Literature Review

Due: 30 October 2026 | Weightage: 3% | Individual | Submission Only

- Each student must individually submit a 1-page literature review covering at least 10–15 papers relevant to their project idea. (Unlike the group-based Idea Presentation, this is an individual deliverable.)
- The review should demonstrate an understanding of relevant research and existing approaches rather than simply summarizing papers.
- AI tools should not be used to write or summarize the literature review. They may only be used for finding relevant papers.
- For each paper, explain its relevance to your own project idea, specifically:
    - How does the paper differ from your proposed project?
    - How does it complement or inform your proposed approach?
    - The final paragraph of the literature review should clearly articulate the novel aspects of your project idea in light of the existing related work.

Each student must submit their individual literature review document on LMS by **30 October 2026**.

The **Idea Presentation** is a group assignment focused on proposing a project idea, identifying a preliminary research gap, and explaining the proposed solution, methodology, and expected impact.

The **Literature Review** is an individual assignment requiring a 1-page critical analysis of 10–15 relevant research papers. Students must compare existing approaches, identify limitations, and explain how the literature informs their proposed project and establishes its novelty.

In short, the Idea Presentation focuses on **what you propose and why**. The Literature Review examines **what has already been done and how your work differs**.

3. Refined Idea Presentation

Each group must submit a refined version of its original Idea Presentation, incorporating insights from the individual literature reviews and feedback received during the initial presentation. The objective is to demonstrate how the group's understanding of existing research has improved and how this has influenced the proposed project.

The refined presentation should cover:

1. **Updated Motivation and Research Gap:** Refine the original problem statement and research gap based on findings from the literature review. Clearly identify the limitations of existing approaches.
2. **Refined Solution and Novelty:** Explain how the proposed solution differs from existing research and what makes the approach novel. Address any weaknesses identified in the original proposal.
3. **Dataset and Methodology:** Present the updated dataset selection, technical approach, and experimental methodology, incorporating relevant insights from existing research.
4. **Expected Results and Evaluation:** Identify suitable evaluation metrics, relevant baselines, and expected outcomes based on the reviewed literature.
5. **Impact and Feasibility:** Explain the potential contribution of the proposed work and demonstrate that the project can reasonably be completed within the available timeframe.

Submission Guidelines

- The presentation must contain **exactly 5 slides**, following the structure above.
- This is a **group submission only**. No in-class presentation is required.
- The submission should reflect meaningful improvements over the original Idea Presentation rather than simply reproducing the same slides.
- Changes should be supported by findings from the literature review, with appropriate references where relevant.
- Submit the refined presentation in **PDF or PPTX format**.

Groups must submit their refined presentation slides on LMS by **31 October 2026**.

4. Short Paper

Due: 15 November 2026 | Weightage: 5% | Group | Submission Only

Each group must submit a short research paper documenting the progress of its project.

The paper should present the problem, relevant literature, proposed methodology, and progress made toward implementing and evaluating the solution.

The submission should include:

- **Abstract:** A concise summary of the problem, proposed approach, and current progress.
- **Introduction:** Background, motivation, and problem statement.
- **Related Work:** Relevant research and how the proposed approach differs from or builds upon existing work.
- **Dataset:**
- **Methodology:** Proposed system design, techniques, and evaluation strategy.
- **Implementation Progress:** Work completed so far, including technical decisions and any challenges encountered.
- **Preliminary Results:** Initial experiments or findings, where available.
- **Remaining Work:** Planned experiments, improvements, and steps toward project completion.
- **References:** Properly cited academic and technical sources.

The paper should demonstrate meaningful project progress and present a clear plan for completing the remaining work.

Groups must submit their short paper on LMS by **15 November 2026**.

5. Full Project Presentation

Due: 22 November 2026 | Weightage: 3% | Group | Presentation

Each group must present its completed project, focusing on the implemented methodology, experimental results, and findings.

Unlike the initial Idea Presentation, this presentation should emphasize **what was actually implemented and evaluated**, rather than what was originally proposed.

The presentation should cover:

1. **Problem and Research Gap:** Briefly introduce the research problem, motivation, and limitations of existing approaches.
2. **Proposed Solution and Contributions:** Explain the final solution, its novelty, and the specific contributions of the project.
3. **Dataset and Methodology:** Describe the datasets, preprocessing steps, model or system architecture, and implementation details. Explain any significant changes from the original proposal.
4. **Experimental Setup:** Present the evaluation methodology, baseline approaches, performance metrics, and experimental configurations.
5. **Results and Analysis:** Present the experimental findings using appropriate tables, figures, or visualizations. Compare results against relevant baselines and explain the observed outcomes.
6. **Limitations and Discussion:** Discuss the limitations of the approach, challenges encountered, and possible reasons for unexpected or unsuccessful results.
7. **Conclusion and Future Work:** Summarize the findings, contributions, and potential directions for improvement.

Presentation Guidelines

- The presentation should focus primarily on the methodology, experiments, results, and critical analysis.

- Results must be supported by experimental evidence. Unsupported claims about model or system performance should be avoided.
- Groups should clearly distinguish between the originally proposed methodology and what was actually implemented.
- All group members should understand the project's technical details and be prepared to answer questions.
- Slides should be concise, readable, and supported by relevant technical diagrams, tables, and figures.

Groups will present their completed projects during the scheduled session on **22 November 2026**.

6. Full Project Paper

Due: 30 November 2026 | Weightage: 20% | Group | Submission Only

Each group must submit a final research paper documenting its completed project. The final paper should be written and formatted as a conference-style research paper.

Conference Selection and Formatting Requirements

- Each group must select a conference relevant to its project topic that has a **CORE A or A*** ranking.
- Conference rankings can be checked at: https://portal.core.edu.au/conf-ranks/
- The paper must follow official submission format and guidelines of selected conference.
- For machine learning and AI projects, relevant conferences may include **NeurIPS, ICML, ICLR, and AAAI**.
- For software engineering projects, relevant conferences may include **ICSE and FSE**.
- Groups must follow the selected conference's paper template, formatting requirements, page limits, and reference style.

Paper Content

The final paper may include the following components, organized according to the selected conference's guidelines:

1. **Abstract:** Concisely summarize the research problem, proposed approach, main experimental findings, and contributions.
2. **Introduction:** Explain the problem, motivation, research gap, objectives, and contributions of the project.
3. **Related Work:** Critically discuss relevant existing research, compare previous approaches, and establish how the project differs from or extends prior work.
4. **Dataset (If relevant):** Describe the datasets used, their sources, characteristics, preprocessing, and any relevant limitations.
5. **Methodology:** Provide a detailed technical explanation of the proposed approach, including models, algorithms, system architecture, and key design decisions.

6. **Experimental Setup:** Describe the implementation environment, experimental configurations, evaluation metrics, baselines, and procedures needed to assess the approach.
7. **Results and Analysis:** Present the experimental findings through appropriate tables and figures. Compare performance against relevant baselines and critically analyze the results.
8. **Discussion and Limitations:** Discuss the implications of the findings, limitations of the methodology, unsuccessful experiments, and possible threats to validity.
9. **Conclusion and Future Work:** Summarize the main contributions, findings, and potential extensions.
10. **References:** Include properly formatted citations for all referenced research and technical resources.

A good starting point would be to look at format and writing of similar conference papers & your reference papers.

AI tools should not be used to write final paper.

Submission Guidelines

- The paper must follow the official template of the selected **CORE A/A*** conference.
- The submission must accurately reflect the work completed by the group.
- Experimental results should be reproducible where reasonably possible, with sufficient methodological and experimental details.
- Claims about novelty, effectiveness, or improvements over existing approaches must be supported by relevant literature or experimental evidence.
- Negative or inconclusive results are acceptable if they are properly reported and critically analyzed.
- The final paper must clearly distinguish between proposed methods and implemented work.

Groups must submit their completed research paper on LMS by **30 November 2026**.